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Greater China Semiconductors | Asia Pacific

Old Memory: Upside Surprise Ahead

We think wider supply/demand will boost earnings. SLC NAND and Si Cap should offer re-rating potential. We upgrade Winbond and Nanya to OW.

Key Takeaways

- We reduce our supply assumptions from Micron in the near term and for Hynix in the next few quarters.
- We believe DDR4 enterprise demand from both eSSD and datacenters is getting stronger.
- Our DDR4 supply/demand model suggests a 19-20% gap for 2H26 vs. the prior 14%, and an 18-20% gap for 2027-28.
- SiCap and SLC NAND (for servers) could be a potential re-rating factor.
- NOR is seeing more tightness on mature foundry shortages.

We had been too conservative; supply/demand gap widening in 2H: We downgraded Winbond and Nanya in March 2026, citing a potentially narrowing supply/demand gap into 2H26 and moderating pricing hikes. We now believe the gap could widen into 2H. We also expect rising CPU server demand to drive enterprise demand growth into 2027. As a result, we upgrade both stocks back to OW.

Less supply and rising demand = stronger pricing: We believe the recent Micron (covered by Joseph Moore) announcement ([link](#)) will mean reduced supply in the next 2-4 quarters, reflecting equipment movement from Taiwan to the US. At the same time, we think Hynix's (covered by Shawn Kim) Wuxi capacity for D4/LPD4 will be further reduced for DDR5 back-end capacity. We thus expect DDR4 pricing to continue to trend up, likely 20% in 3Q. Our supply/demand models suggests a 19-20% gap in 2H26 and an 18-20% gap in 2027/28.

SiCap and SLC NAND could be re-rating factors: SiCap ([link](#)) is a new growth driver for niche memory vendors. Recent developments for servers (for high performance NAND) also suggest potential adoption of SLC in datacenters, as SLC is ideal for fast reading and writing speed. Currently, SLC NAND could be supplied by Greater China vendors such as GigaDevice, Winbond, and Macronix.

Estimate changes: For **Winbond**, we raise 2026, 2027 and 2028 EPS estimates by 23%, 65% and 94%, respectively, to reflect DDR4 pricing strength and opportunities from SiCap and SLC NAND. For **Nanya**, we increase respective estimates by 15%, 29% and 33% to reflect DDR4 pricing strength; we also turn positive on more LTAs with server and enterprise SSD customers, along with their additional 5kwpn capacity to DDR5 starting 3Q26. For **GigaDevice**, we raise 2026e by 58% and 2027e by 60% to reflect DDR4 pricing strength and opportunities from SLC NAND.

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GREATER CHINA TECHNOLOGY SEMICONDUCTORS

Asia Pacific

Industry View

Attractive

WHAT'S CHANGED

Nanya Technology Corp.
(2408.TW)

Rating

Price Target

From

Equal-weight

NT\$278.00

To

Overweight

NT\$380.00

Winbond Electronics Corp
(2344.TW)

Rating

Price Target

From

Equal-weight

NT\$100.00

To

Overweight

NT\$222.00

GigaDevice Semiconductor
Beijing Inc (603986.SS)

Price Target

From

Rmb348.00

To

Rmb585.00

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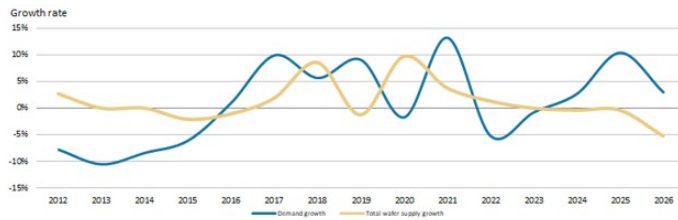
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Key Charts and Exhibits

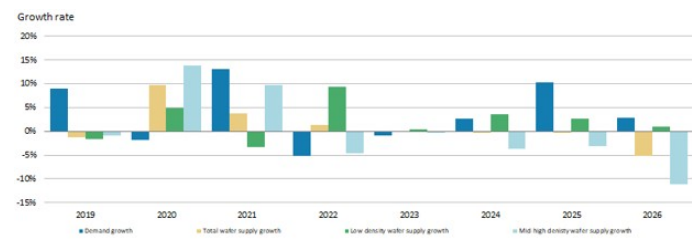
Exhibit 1: Order of preference: GC memory

	Macronix 2337.TW	AP Memory 6531.TW	Winbond 2344.TW	Nanya Tech 2408.TW	PSMC 6770.TW	GigaDevice 603986.SS
						
Rating	Overweight	Overweight	Overweight	Overweight	Overweight	Overweight
Trading Currency	TWD	TWD	TWD	TWD	TWD	CNY
Price Target	202.0	1,555.0	222.0	380.0	88.0	585.0
Current Price (as of 2026/5/27)	155.5	1,080.0	155.0	312.0	74.8	514.2
Upside/(Downside) (%)	30%	44%	43%	22%	18%	14%
Market Cap (in USD mm)	8,932.0	5,562.4	20,620.6	30,605.2	9,970.0	50,574.4
Avg Daily Traded Vol (in USD mm)	215.3	68.1	372.8	558.3	206.4	1,088.1
Street View: Ratings						
Buy/Overweight	90%	75%	91%	75%	50%	93%
Hold/Equal-weight	0%	25%	9%	19%	50%	7%
Sell/Underweight	10%	0%	0%	6%	0%	0%
Bull Case Value	326.0	1,830.0	264.0	805.0	140.0	1,016.0
Upside (%)	110%	69%	70%	158%	87%	98%
Bear Case Value	119.5	620.0	91.0	210.0	44.0	427.0
Downside (%)	-23%	-43%	-41%	-33%	-41%	-17%
Risk/Reward Skew	4.7	1.6	1.7	4.8	2.1	5.8
Morgan Stanley Estimates						
FY26e	TWD	TWD	TWD	TWD	TWD	CNY
Sales	66,682	9,601	258,581	332,719	73,014	22,018
EBITDA	24,507	3,037	129,462	253,840	13,009	9,882
EBIT	16,970	2,976	113,852	241,654	5,445	9,632
EPS	7.35	17.65	20.36	61.50	4.64	12.81
FY27e						
Sales	126,013	17,829	377,335	471,750	89,835	32,133
EBITDA	54,966	6,491	203,533	342,723	27,573	15,564
EBIT	48,496	6,429	183,356	319,538	20,109	15,314
EPS	20.82	34.34	32.73	80.92	3.78	20.71
FY26 MSe vs. Consensus Mean						
Sales	-16.5%	-3.0%	11.7%	13.0%	6.7%	24.5%
EBITDA	-22.7%	-0.9%	12.3%	13.5%	-55.9%	42.6%
EBIT	-40.8%	3.8%	9.9%	15.0%	-78.0%	47.9%
EPS	-44.7%	-0.7%	9.1%	23.8%	-13.7%	50.6%
FY27 MSe vs. Consensus Mean						
Sales	6.8%	19.6%	25.9%	21.5%	8.8%	41.2%
EBITDA	-18.8%	26.0%	19.5%	16.5%	-9.7%	80.2%
EBIT	-13.2%	33.0%	37.9%	14.5%	-4.7%	81.6%
EPS	-24.8%	25.1%	31.5%	24.8%	-8.3%	91.6%
Valuation Multiples at Last Close						
FY26e						
P/E	21.1x	61.2x	7.6x	5.1x	16.1x	40.1x
EV/EBIT	24.2x	80.6x	8.5x	5.6x	34.0x	38.9x
EV/EBITDA	13.3x	53.6x	5.3x	4.5x	13.9x	33.1x
EV/Sales	4.9x	17.0x	2.7x	3.4x	2.5x	14.9x
FCF Yield	-11.6%	1.2%	4.7%	-23.4%	32.4%	2.1%
FY27e						
P/E	7.5x	31.4x	4.7x	3.9x	19.8x	24.8x
EV/EBIT	5.9x	25.0x	2.9x	2.8x	6.7x	20.7x
EV/EBITDA	5.2x	24.8x	2.6x	2.6x	4.9x	20.4x
EV/Sales	2.3x	9.0x	1.4x	1.9x	1.5x	9.9x
FCF Yield	15.3%	2.2%	24.5%	26.1%	16.0%	3.5%
Implied Multiples on MS Price Target						
FY26e						
P/E	27.5x	88.1x	10.9x	6.2x	19.0x	45.7x
EV/EBIT	24.2x	80.6x	8.5x	5.6x	34.0x	38.9x
EV/EBITDA	16.7x	79.0x	7.5x	5.3x	14.2x	37.9x
EV/Sales	6.1x	25.0x	3.7x	4.1x	2.5x	17.0x
FY27e						
P/E	9.7x	45.3x	6.8x	4.7x	23.3x	28.2x
EV/EBIT	7.7x	40.0x	5.2x	3.8x	18.8x	28.9x
EV/EBITDA	6.8x	39.6x	4.7x	3.5x	13.7x	28.4x
EV/Sales	3.0x	14.4x	2.5x	2.5x	4.2x	13.8x

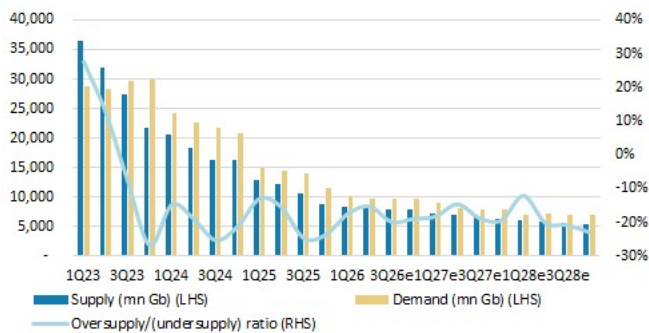
Source: FactSet (consensus mean), Morgan Stanley Research (e) estimates.

Exhibit 2: NOR flash demand and supply growth rates

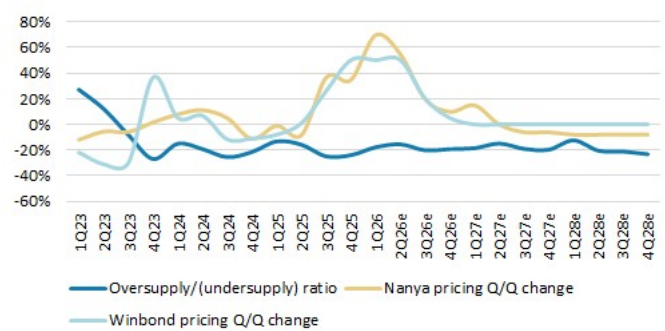
Source: Company data, Morgan Stanley Research

Exhibit 3: NOR flash demand growth and supply growth by density

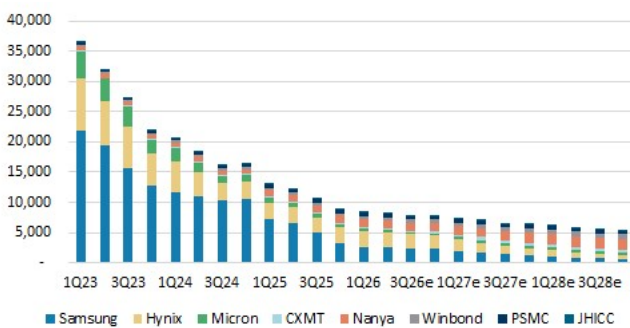
Source: Company data, Morgan Stanley Research

Exhibit 4: DDR4 quarterly supply and demand summary

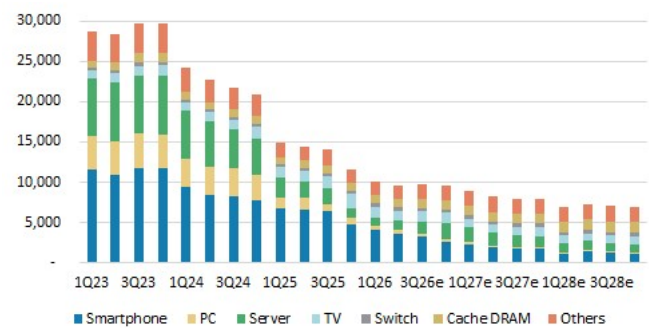
Source: Company data, IDC, Morgan Stanley Research (e) estimates

Exhibit 5: Quarterly oversupply/undersupply ratio vs. Nanya and Winbond pricing Q/Q change

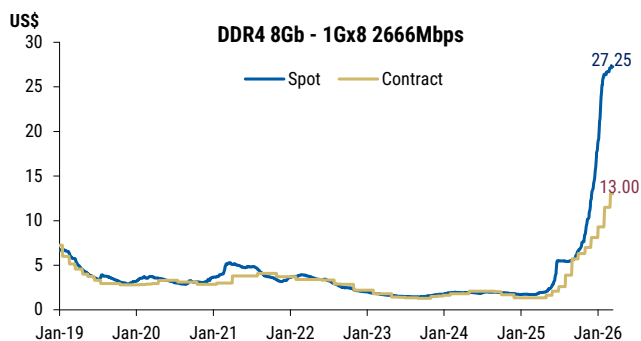
Source: Company data, IDC, Morgan Stanley Research (e) estimates

Exhibit 6: Quarterly supply breakdown (mn Gb)

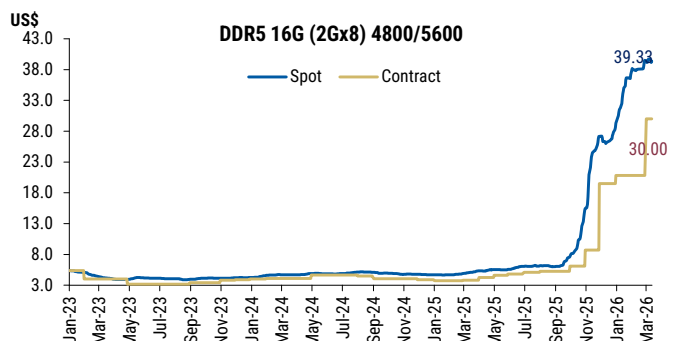
Source: Company data, Morgan Stanley Research (e) estimates

Exhibit 7: Quarterly demand breakdown by product (mn Gb)

Source: IDC, Morgan Stanley Research (e) estimates

Exhibit 8: DDR4 8Gb (1Gx8) pricing chart

Source: DRAMeXchange, Morgan Stanley Research. Data as of Mar 13, 2026.

Exhibit 9: DDR5 16Gb (2Gx8) pricing chart

Source: DRAMeXchange, Morgan Stanley Research. Data as of Mar 13, 2026.

Winbond: Estimate Revisions and Quarterly Financials

We raise our 2026, 2027, and 2028 EPS estimates by 23%, 65%, and 94%, respectively. This reflects DDR4 pricing strength, an enlarged SLC NAND TAM, and SiCap foundry business with SEMCO.

Exhibit 10: Winbond: Estimate revisions

NT\$ mn	New '26E	Old '26E	Diff.	New '27E	Old '27E	Diff.	New '28E	Old '28E	Diff.
Net sales	258,581	224,249	15%	377,335	264,897	42%	410,515	256,015	60%
COGS	(111,702)	(103,295)		(158,768)	(123,965)		(178,012)	(127,994)	
Gross profit	146,879	120,954	21%	218,567	140,932	55%	232,503	128,021	82%
Operating expenses	(33,028)	(30,277)		(35,211)	(31,693)		(36,149)	(31,941)	
Operating profit	113,852	90,677	26%	183,356	109,239	68%	196,354	96,080	104%
Non-op. income (exp)	715	917		776	917		756	897	
Pretax income	114,567	91,594	25%	184,133	110,157	67%	197,111	96,977	103%
Taxes	(22,946)	(18,319)		(36,827)	(22,031)		(40,609)	(20,583)	
Net income	91,607	74,586	23%	147,293	89,436	65%	156,488	77,705	101%
Reported EPS	20.36	16.57	23%	32.73	19.87	65%	36.09	18.59	94%
Margins									
Gross margin	56.8%	53.9%	2.9 ppt	57.9%	53.2%	4.7 ppt	56.6%	50.0%	6.6 ppt
Operating margin	44.0%	40.4%	3.6 ppt	48.6%	41.2%	7.4 ppt	47.8%	37.5%	10.3 ppt
Pretax margin	44.3%	40.8%	3.5 ppt	48.8%	41.6%	7.2 ppt	48.0%	37.9%	10.1 ppt
Net margin	35.4%	33.3%	2.2 ppt	39.0%	33.8%	5.3 ppt	38.1%	30.4%	7.8 ppt
Opex %	12.8%	13.5%	-0.7 ppt	9.3%	12.0%	-2.6 ppt	8.8%	12.5%	-3.7 ppt
NT\$	New '26E	Old '26E	Diff.	New '27E	Old '27E	Diff.	New '28E	Old '28E	Diff.
BVPS	40.54	37.59	8%	73.32	57.51	27%	82.20	62.39	32%

Source: Morgan Stanley Research (E) estimates

Exhibit 11: Winbond: Quarterly financials

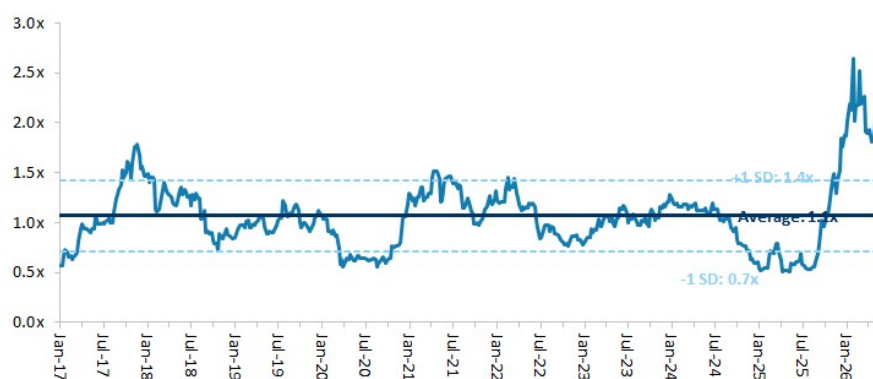
NT\$ in million	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26	4Q26	1Q27E	2Q27E	3Q27E	4Q27E	1Q28E	2Q28E	3Q28E	4Q28E	2025	2026E	2027E	2028E
Total Revenues	19,993	21,019	21,771	26,625	38,253	64,396	79,948	89,984	91,342	93,039	95,199	97,735	100,164	102,137	103,626	104,778	89,406	298,581	377,335	410,515
Sequential Change	7.0%	5.1%	3.6%	22.3%	42.7%	42.2%	39.6%	18.5%	1.5%	1.9%	2.3%	2.7%	2.3%	2.0%	1.3%	1.3%				
Change vs Year Ago	-0.6%	-2.2%	2.1%	42.4%	91.3%	158.8%	248.9%	236.0%	138.8%	71.0%	25.3%	8.6%	9.7%	9.8%	6.7%	7.2%	9.6%	189.2%	45.9%	8.8%
Cost of Sales	(14,871)	(16,255)	(11,605)	(15,479)	(17,830)	(23,585)	(31,990)	(37,885)	(38,034)	(38,887)	(40,252)	(41,025)	(41,404)	(42,428)	(43,627)	(44,617)	(58,210)	(111,702)	(158,768)	(178,012)
Percent of Revenues	74.4%	77.3%	53.3%	58.1%	46.6%	44.1%	42.1%	42.1%	41.6%	41.8%	42.3%	42.6%	41.3%	41.5%	42.2%	42.6%	65.1%	42.2%	42.1%	43.4%
Gross Margin	5,122	4,763	10,166	11,146	20,415	30,407	43,957	52,100	53,318	54,172	54,947	56,130	58,760	60,709	59,999	60,160	31,194	146,879	218,567	232,503
Percent of Revenues	25.6%	22.7%	46.7%	41.9%	53.4%	53.9%	55.9%	57.9%	58.4%	59.2%	57.7%	57.4%	58.7%	58.5%	57.8%	57.4%	34.9%	56.8%	57.5%	56.6%
Total Opex	(6,087)	(6,058)	(6,463)	(7,054)	(7,055)	(7,919)	(8,463)	(8,780)	(8,565)	(8,677)	(8,912)	(9,050)	(9,040)	(9,040)	(9,031)	(9,012)	(25,062)	(33,028)	(35,211)	(36,149)
Percent of Revenues	30.4%	28.8%	29.7%	26.5%	20.6%	14.6%	11.1%	9.8%	9.4%	9.3%	9.4%	9.3%	9.0%	8.9%	8.7%	8.6%	28.7%	12.8%	9.3%	8.8%
Operating Income	(665)	(1,295)	3,761	4,062	12,550	22,488	39,464	43,319	44,753	45,464	46,025	47,074	49,734	50,660	50,779	51,148	5,534	113,852	183,356	196,354
Percent of Revenues	-4.8%	-6.2%	17.0%	15.4%	32.8%	41.3%	46.7%	48.1%	49.0%	48.9%	48.4%	48.2%	49.6%	49.6%	49.1%	48.8%	6.2%	44.0%	48.6%	47.8%
Total Non-operating Income(Loss)	(180)	(524)	(145)	(75)	138	199	189	189	199	199	189	189	189	189	189	189	(924)	715	776	756
Percent of Revenues	-0.9%	-2.5%	-0.7%	-0.3%	0.4%	0.3%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	-1.0%	0.3%	0.3%	0.3%
Profit Before Taxes	(1,140)	(1,819)	3,558	4,017	12,688	22,687	35,653	43,508	44,952	45,664	46,225	47,263	49,923	50,849	50,968	51,337	4,610	114,567	184,133	197,111
Percent of Revenues	-5.7%	-8.7%	16.3%	15.1%	33.2%	41.7%	47.0%	48.4%	49.2%	49.1%	48.8%	48.7%	49.8%	49.8%	49.3%	49.0%	5.2%	44.3%	48.8%	47.8%
Taxes	159	188	(857)	(922)	(4,537)	(7,137)	(8,702)	(8,990)	(8,139)	(8,245)	(8,453)	(8,579)	(10,170)	(10,194)	(10,267)	(10,267)	(1,433)	(22,946)	(36,827)	(40,609)
Tax Rate	13.9%	10.3%	-24.1%	-23.0%	-20.3%	-20.0%	-20.0%	-20.0%	-20.0%	-20.0%	-20.0%	-20.0%	-20.0%	-20.0%	-20.0%	-20.0%	31.1%	20.0%	20.0%	20.0%
Reported Income (TW GAAP)	(1,091)	(1,617)	2,693	3,095	10,144	18,146	28,543	34,802	36,818	36,818	37,772	38,707	39,714	40,676	40,711	41,066	3,177	91,607	147,293	156,488
Percent of Revenues	-5.5%	-7.7%	12.3%	11.6%	26.4%	33.4%	37.6%	38.7%	39.4%	39.3%	39.3%	39.3%	39.3%	39.3%	39.3%	39.3%	3.6%	35.4%	38.0%	36.2%
Change vs Year Ago	135.0%	N/A	N/A	N/A	0.0%	N/A	N/A	255.5%	101.4%	29.5%	8.6%	11.0%	11.3%	10.2%	8.6%	559.2%	2212.2%	60.6%	6.2%	
Reported EPS (NT\$ TW GAAP)	(8.34)	(8.29)	0.65	0.76	2.25	4.63	8.34	7.73	7.99	8.12	8.22	8.40	8.67	8.68	8.68	8.68	0.88	20.36	32.73	36.09
Sequential Change	61%	20%	324%	6%	158%	79%	57%	22%	7%	7%	7%	7%	8%	7%	0%	1%	493.9%	2212.2%	60.6%	10.3%
Change vs Year Ago	118%	N/A	N/A	N/A	0%	870%	917%	256%	101%	30%	9%	11%	11%	10%	9%	9%				
Book value per share	21.8	21.0	22.5	25.4	28.5	30.5	32.8	40.5	48.5	56.7	64.9	73.3	82.2	90.0	93.8	93.4	25.4	40.5	73.3	82.2

Source: Company data, Morgan Stanley Research (E) estimates

Winbond: Valuation Methodology

We raise our PT to NT\$222 from NT\$100: We continue to apply P/B multiple methodology to derive our price target, in line with our approach for Greater China memory IDM peers and in view of the industry's high earnings volatility. Our new price target (base case scenario value) implies 5.5x 2026e BVPS and 3.0x 2027e BVPS, vs. our prior target multiple of 2.7x 2026e BVPS, as we think DDR4 pricing and strength and SLC NAND / SiCap opportunities could drive re-rating. We raise our bull case value to NT\$264 (6.5x our 2026e BVPS) and our bear case value to NT\$91 (2.2x our 2026e BVPS).

Exhibit 12: Winbond: Historical forward P/B



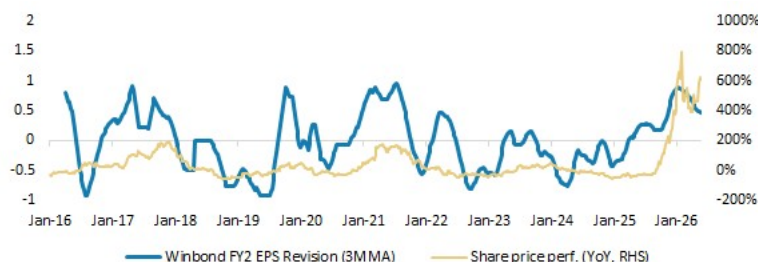
Source: Company data, Morgan Stanley Research estimates

Exhibit 13: Winbond: Historical forward P/B vs ROE



Source: Company data, Morgan Stanley Research estimates

Exhibit 14: Winbond: Earnings estimate revision breadth



Source: Company data, Morgan Stanley Research estimates

Risk Reward – Winbond Electronics Corp (2344.TW)

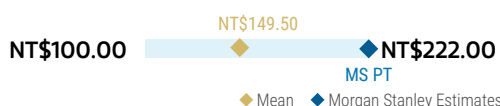
DDR, NOR and SLC NAND pricing upside; long-term opportunities in CUBE

PRICE TARGET NT\$222.00

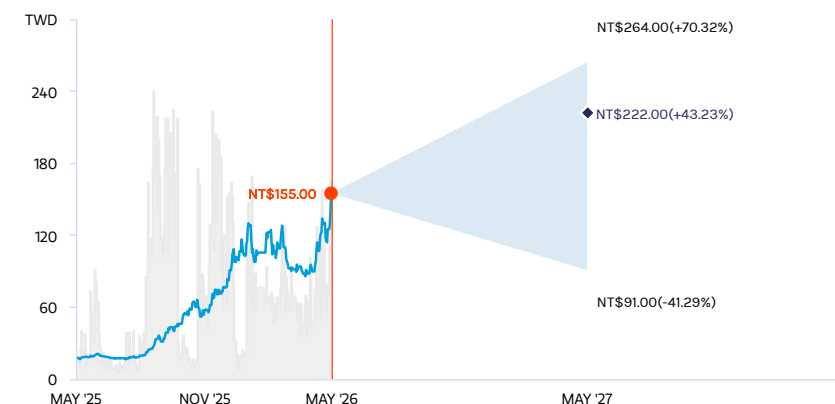
Base case, 5.5x 2026e P/B. This method is in line with the approach we adopt for Greater China memory IDM peers, in view of the industry's high volatility. We believe the stock merits a multiple beyond the high end of its historical band of 0.5-1.5x since 2017, as we are seeing DRAM pricing upside, a sustainable NOR price hike, and opportunities in SLC NAND and SiCap foundry business.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



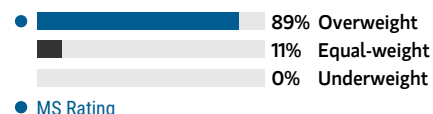
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We expect DDR3 and DDR4 price increases to continue in 2026. The supply/demand gap should widen into 2H26.
- We see NOR price hikes continuing throughout 2026.
- High density SLC NAND pricing is catching up, and we now expect a larger TAM from data center applications.
- We share management's view that CUBE could be meaningful in 2026, given engagement with multiple foundry partners and customers.
- We see potential for several headwinds ahead for the logic business.
- Our PT implies 3.0x 2027e P/B.

Consensus Rating Distribution



● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

NT\$264.00

6.5x 2026e BVPS

NOR Flash pricing and DRAM prices rebound more in the next 12 months: NOR Flash rises dramatically. DRAM prices continue to rise drastically within the next 12 months, with 25nm accounting for >80% of sales.

BASE CASE

NT\$222.00

5.5x 2026e BVPS

Margin expansion benefiting from the memory up-cycle: 1) NOR Flash pricing rises significantly in the next 12 months; and 2) DRAM prices rise significantly in the next 12 months.

BEAR CASE

NT\$91.00

2.2x 2026e BVPS

NOR Flash prices face pressure in the next 12 months; DRAM market back to oversupply, with slower technology migration: 1) NOR Flash shows price declines; and 2) DRAM prices drop in the next 12 months.

Risk Reward – Winbond Electronics Corp (2344.TW)

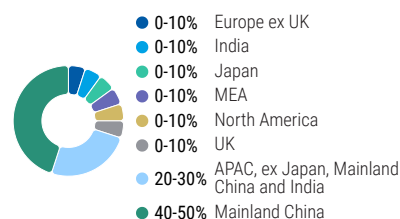
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
NOR ASP (%)	8	91	15	(9)
DRAM ASP (%)	(2)	250	23	5
Mobile RAM sales (NT\$, mn)	5,302	20,466	31,117	33,443
DRAM sales (NT\$, mn)	27,808	162,393	262,521	295,448
Flash sales (NT\$, mn)	31,106	65,008	83,222	76,142

INVESTMENT DRIVERS

- DRAM pricing
- NOR pricing
- Technology migration

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

1/5
MOST 3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- NOR Flash pricing increase, driven by stronger-than-expected demand
- Stronger-than-expected DRAM pricing given ongoing supply shortage
- Faster-than-expected SLC NAND development

RISKS TO DOWNSIDE

- NOR Flash downcycle, driven by weaker demand
- Lower-than-expected DRAM pricing given ongoing oversupply
- Slower-than-expected SLC NAND development

OWNERSHIP POSITIONING

Inst. Owners, % Active 47.5%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (NT\$, mn) 187,233 258,581 274,712
231,525

EBITDA (NT\$, mn) 77,402 129,462 156,063
115,321

Net income (NT\$, mn) 52,797 91,608 114,946
84,043

EPS (NT\$) 11.73 20.36 25.54
18.66

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Winbond: Financial Summary

Exhibit 15: Financial summary

Income Statement

NT\$m (Years End Dec)	2025	2026E	2027E	2028E
Net sales	89,406	258,581	377,335	410,515
COGS	(58,210)	(111,702)	(158,768)	(178,012)
Depreciation Expense	(12,083)	(15,219)	(19,785)	(25,721)
Variable Cost	(26,224)	(77,544)	(120,048)	(129,023)
Gross profit	31,196	146,879	218,567	232,503
Operating expenses	(25,662)	(33,028)	(35,211)	(36,149)
Operating income	5,534	113,852	183,356	196,354
Non-operating income	(924)	715	776	756
Pre-tax income	4,610	114,567	184,133	197,111
Income tax	(649)	(22,960)	(36,840)	(40,623)
Reported net income	3,962	91,607	147,293	156,488
Adj. wtd. avg. shrs (m)	4,500	4,500	4,500	4,500
Reported EPS (NT\$)	0.88	20.36	32.73	36.09
Modelware EPS (NT\$)	0.88	20.36	32.73	36.09

Balance Sheet

NT\$m (Years End Dec)	2025	2026E	2027E	2028E
Cash	15,734	19,403	169,426	207,048
Mkt Securities	13,347	13,347	13,347	13,347
AR/NR	16,071	51,991	56,480	57,873
Inventory	25,758	96,884	105,250	107,844
Other	2,281	2,281	2,281	2,281
Current Assets	73,191	183,906	346,785	388,393
Long-term investments	19,266	19,216	19,156	19,146
Fixed assets	93,800	88,581	78,796	76,838
Other assets	5,935	5,935	5,935	5,935
Total Assets	192,192	297,638	450,672	490,312
S/T borrowings	2,739	2,739	2,739	2,739
AP/NP	8,224	26,133	28,713	28,561
Other ST liabilities	35,652	54,893	57,832	57,658
LT debt	26,399	26,399	26,399	26,399
Other LT liabilities	5,048	5,048	5,048	5,048
Total Liabilities	78,062	115,213	120,731	120,406
Common stocks	45,000	45,000	45,000	45,000
Preferred stocks	0	0	0	0
Capital Reserve	13,752	13,752	13,752	13,752
Retained earning	49,184	117,480	264,994	304,961
Other shareholders' equity	0	0	0	0
Total Equity	114,130	182,426	329,940	369,907
Total Liab. & Shrhldr's Equity	192,192	297,638	450,672	490,312
BVPS	25.4	40.5	73.3	82.2

Cash Flow Statement

NT\$m (Years End Dec)	2025	2026E	2027E	2028E
Cashflow from Operations	18,668	45,959	168,780	180,980
Net profits	3,962	91,607	147,293	156,488
Depreciation	12,308	15,219	19,785	19,785
Working Capital Change	(6,405)	(89,136)	(10,276)	(4,139)
Other adjustments	8,802	28,269	11,978	8,845
Cashflow from Investing	(18,259)	(10,342)	(10,331)	(18,209)
Capex	(5,300)	(10,000)	(10,000)	(10,000)
Change of LT Investment	0	0	0	0
Change of ST Investment	(4,827)	0	0	0
Other adjustments	(8,132)	(342)	(331)	(8,209)
Cashflow from financing	9,931	(23,311)	222	(116,522)
Increase in L/T debt	(1,698)	0	0	0
Increase in S/T debt	619	0	0	0
Cash Dividend Paid	0	(18,321)	0	0
Dir & Emp Bonus Paid	0	0	0	0
Other adjustments	11,009	(4,990)	222	(116,522)
Exchange rate adjustment	(8,706)	(8,636)	(8,647)	(8,627)
Net change in cash	1,633	3,669	150,023	37,622

Financial Ratio

	2025	2026E	2027E	2028E
Growth (%)				
Turnover	9.6	189.2	45.9	8.8
Margins (%)				
Gross Margin	34.9	56.8	57.9	56.6
Operating Margin	6.2	44.0	48.6	47.8
EBITA Margin	-8.0	38.1	43.3	41.6
Pretax Margin	5.2	44.3	48.8	48.0
Net Profit	4.4	35.4	39.0	38.1
Return (%)				
ROAE	4.3	84.9	83.6	48.3
ROAA	2.1	37.4	39.4	33.3
Gearing (%)				
Net Debt/Equity	31.9	17.9	(35.6)	(41.9)
Liabilities/Equity	68.4	63.2	36.6	32.6
Ratios (X)				
Current ratio	1.6	2.2	3.9	4.4
Quick ratio	0.7	0.9	2.5	3.0
Others				
AR/NR Turnover (days)	45	53	48	52
Inventory Turnover (days)	107	102	87	98
AP Turnover (days)	45	52	85	66
Cash Conversion (days)	107	104	49	84

Source: Company data, Morgan Stanley Research (E) estimates

GigaDevice: Estimate Revisions and Quarterly Financials

We raise our EPS estimates by 58% for 2026 and 60% for 2027. This reflects DDR4 pricing strength and an enlarged SLC NAND TAM.

Exhibit 16: GigaDevice: Estimate revisions

Rmb mn	2026E	2026E	Diff.	2027E	2027E	Diff.	2028E
Net sales	22,018	17,061	29%	32,133	24,555	31%	37,996
COGS	9,437	8,509		13,736	12,045		17,583
Gross profit	12,581	8,552	47%	18,397	12,510	47%	20,413
Operating expenses	2,949	2,816		3,083	3,275		3,354
Operating profit	9,632	5,736	68%	15,314	9,235	66%	17,059
Non-op. income (exp.)	(125)	100		20	100		20
Pretax income	9,507	5,836	63%	15,334	9,335	64%	17,079
Taxes	873	175		1,408	280		1,568
Net income	8,589	5,628	53%	13,881	9,022	54%	15,465
Reported EPS (Rmb)	12.81	8.10	58%	20.71	12.98	60%	23.07
Margins							
Gross margin	57.1%	50.1%	7.0%	57.3%	50.9%	6.3%	53.7%
Operating margin	43.7%	33.6%	10.1%	47.7%	37.6%	10.0%	44.9%
Pretax margin	43.2%	34.2%	9.0%	47.7%	38.0%	9.7%	44.9%
Net margin	39.0%	33.0%	6.0%	43.2%	36.7%	6.5%	40.7%

Source: Morgan Stanley Research (E) estimates

Exhibit 17: GigaDevice: Quarterly financials

Rmb mn	1Q21	2Q21	3Q21	4Q21	1Q22	2Q22	3Q22	4Q22	1Q23	2Q23	3Q23	4Q23	1Q24	2Q24	3Q24	4Q24	2024	2025	2026E	2027E	2028E
Total Revenues	1,309	2,241	2,641	2,372	4,188	4,957	6,395	6,316	7,314	7,809	8,303	8,614	8,812	9,374	9,727	10,140	7,266	9,203	22,018	32,133	37,996
Sequential Change	11.9%	17.4%	19.6%	-11.5%	76.6%	18.4%	19.7%	16.6%	5.5%	6.0%	5.1%	3.7%	2.8%	4.8%	4.4%	4.2%	27.7%	25.1%	139.2%	45.9%	18.2%
Change vs Year Ago	17.3%	13.1%	31.4%	39.0%	119.4%	121.2%	121.4%	192.4%	74.7%	59.3%	39.9%	24.2%	21.0%	17.5%	17.1%	17.7%	27.7%	25.1%	139.2%	45.9%	18.2%
Cost of Sales	1,194	1,412	1,580	1,367	1,796	2,276	2,558	2,803	3,010	3,348	3,610	3,761	3,945	4,255	4,539	4,805	4,581	5,502	9,437	13,736	17,583
Percent of Revenues	63%	63%	59%	58%	43%	46%	40%	44%	41%	42%	44%	44%	45%	46%	47%	47%	63%	60%	43%	42%	46%
Gross Profit	715	830	1,062	1,005	2,390	2,680	3,378	4,132	4,308	4,462	4,888	4,853	4,907	5,023	5,148	5,335	2,785	3,701	12,581	18,397	20,413
Percent of Revenues	37.4%	37.0%	40.1%	42.4%	57.1%	54.1%	59.0%	59.0%	57.6%	56.4%	58.3%	55.4%	54.1%	52.9%	52.6%	52.6%	38.0%	40.2%	57.1%	57.3%	53.7%
Incremental Margin	72%	80%	86%	84%	73%	38%	71%	73%	40%	42%	33%	34%	22%	27%	29%	40%	51%	49%	69%	58%	34%
Total Opex	540	545	574	518	660	706	757	819	753	786	820	789	810	852	895	938	1,984	2,176	2,949	3,083	3,354
R&D	292	270	292	257	368	368	383	403	296	300	315	335	290	300	315	335	1,122	1,117	1,513	1,240	1,240
Percent of Revenues	28.2%	24.2%	27.4%	21.6%	15.9%	14.2%	12.8%	11.8%	9.8%	9.5%	9.5%	8.6%	8.9%	8.6%	8.8%	8.8%	27.0%	23.6%	13.4%	9.6%	8.8%
SG&A	150	150	161	141	179	184	194	194	195	200	205	215	216	221	220	230	401	612	732	816	900
Percent of Revenues	11.5%	12.3%	10.9%	10.8%	8.6%	7.4%	6.5%	5.8%	4.0%	3.8%	3.8%	3.9%	3.2%	3.2%	3.2%	3.3%	15.3%	12.1%	6.9%	3.9%	3.9%
General & administrative	136	150	161	141	179	184	194	194	195	200	205	215	216	221	220	230	401	612	732	816	900
Percent of Revenues	10.4%	10.9%	10.9%	10.9%	10.9%	10.9%	10.9%	10.9%	10.9%	10.9%	10.9%	10.9%	10.9%	10.9%	10.9%	10.9%	10.9%	10.9%	10.9%	10.9%	10.9%
Selling & marketing	112	113	121	101	134	158	190	190	224	252	265	275	283	290	311	324	371	448	704	1,027	1,214
Percent of Revenues	8.6%	5.0%	4.5%	4.3%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	5.0%	4.8%	3.2%	3.2%	3.2%
Operating Income	116	285	618	547	1,214	1,874	2,821	3,213	3,587	3,799	3,960	4,028	4,118	4,268	4,298	4,440	811	1,226	9,632	15,314	17,059
Percent of Revenues	9.2%	12.7%	19.3%	23.1%	41.2%	36.8%	44.1%	47.8%	49.0%	48.1%	47.8%	46.8%	46.8%	45.3%	44.2%	43.8%	11.0%	16.0%	43.7%	47.7%	44.9%
Change vs Year Ago	12.7%	19.4%	36.9%	133.9%	883.7%	582.4%	408.0%	505.4%	108.1%	92.5%	48.8%	21.6%	14.8%	10.7%	10.2%	10.2%	130%	88%	531%	59%	11%
Total Non-operating Income(Loss)	60	75	10	35	(103)	(7)	(7)	(7)	5	5	5	5	5	5	5	5	313	181	(120)	20	20
Profit Before Taxes	236	360	628	582	1,611	1,966	2,814	3,206	3,592	3,804	3,965	4,033	4,123	4,210	4,301	4,445	1,124	1,706	9,507	15,334	17,079
Percent of Revenues	12%	16%	20%	25%	39%	40%	44%	48%	49%	49%	47%	47%	47%	45%	44%	44%	15%	19%	43%	48%	45%
Taxes	(4)	(4)	(1)	(1)	(14)	(14)	(14)	(14)	(14)	(14)	(14)	(14)	(14)	(14)	(14)	(14)	(14)	(14)	(14)	(14)	(14)
Net Income	240	348	627	573	1,497	1,780	2,314	3,002	3,262	3,455	3,547	3,663	3,744	3,824	3,900	4,030	1,110	1,677	9,335	15,050	16,965
Percent of Revenues	12%	15%	19%	24%	36%	36%	40%	43%	42%	44%	43%	42%	42%	41%	40%	40%	15%	18%	42%	47%	45%
Change vs Year Ago	10%	9%	61%	109%	523%	421%	365%	430%	122%	84%	50%	22%	15%	11%	10%	10%	584%	49%	421%	62%	11%
Reported EPS (Rmb)	0.36	0.81	0.76	0.88	2.18	2.46	3.82	4.46	4.88	5.14	5.37	5.48	5.67	5.69	5.81	6.01	1.86	2.48	12.81	20.71	23.07
Change vs Year Ago	15%	10%	61%	108%	517%	418%	362%	428%	122%	84%	50%	22%	15%	11%	10%	10%	580%	50%	417%	62%	11%

Source: Company data, Morgan Stanley Research (E) estimates

GigaDevice: Valuation methodology

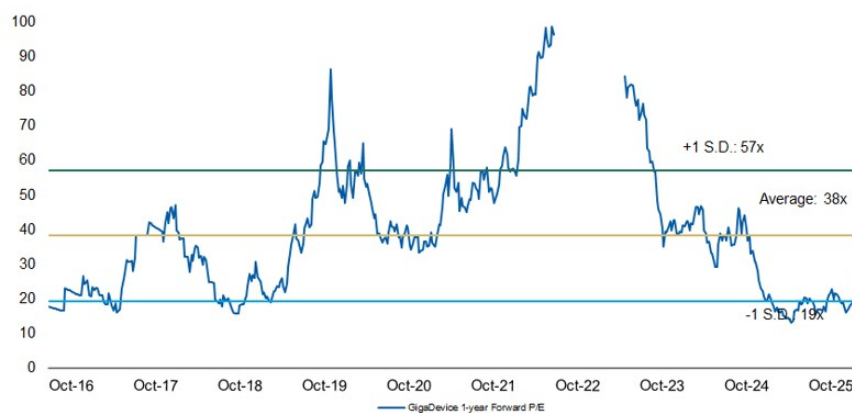
We raise our price target (base case scenario value) by 68% to Rmb585: We continue to derive our price target from a residual income model, which we believe best captures the stock's long-term value. Our key assumptions include a cost of equity of 8.9% (beta 1.1, risk-free rate 2%, and risk premium 6%), a payout ratio of 40%, a medium-term growth rate of 16.4% and a terminal growth rate of 3.0% (all unchanged). Our new price target implies 2026e P/E of 46x, above the historical average of 40x since 2016. Our bull case value rises to Rmb1,016 from Rmb604, and our bear case value increases to Rmb427 from Rmb254.

Exhibit 18: GigaDevice: Residual Income model

Rmb mn	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E
Total Equity	27,317	38,622	49,923	61,407	74,768	90,314	108,402	129,447	153,933	182,423	215,570	254,138
Net Profit	8,589	13,881	15,465	17,994	20,936	24,359	28,342	32,976	38,367	44,640	51,939	60,431
ROAE	36.9%	42.1%	34.9%	32.3%	30.7%	29.5%	28.5%	27.7%	27.1%	26.5%	26.1%	25.7%
Residual Income	5,387	9,074	10,059	11,701	13,425	15,421	17,736	20,425	23,548	27,179	31,402	36,312
Spread	28.0%	33.2%	26.0%	23.4%	21.9%	20.6%	19.6%	18.8%	18.2%	17.7%	17.2%	16.8%
Ending Equity Capital	27,317											
PV of Forecast Period	93,912											
PV of Continuing Value	271,171											
Equity Value	392,401											
No. of Shares	670											
Projected Price (Rmb)	585											

Source: Morgan Stanley Research (E) estimates

Exhibit 19: GigaDevice: historical forward P/E



Source: Company data, Morgan Stanley Research estimates

Risk Reward – GigaDevice Semiconductor Beijing Inc (603986.SS)

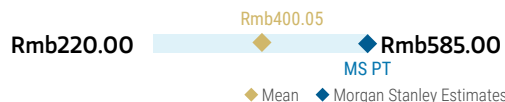
Multiple drivers ahead

PRICE TARGET Rmb585.00

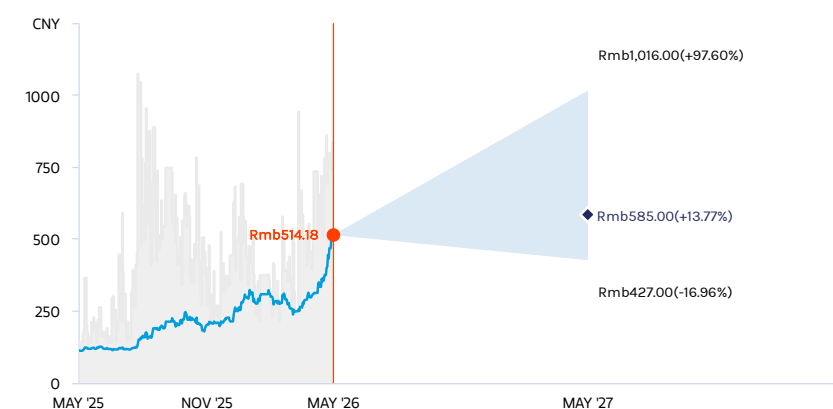
Our price target is our base case value, derived from our residual income model. Key assumptions: cost of equity of 8.9% (beta 1.1, risk-free rate 2% and risk premium 6%), a payout ratio of 40%, medium-term growth rate of 16.4%, and a terminal growth rate of 3.0%.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



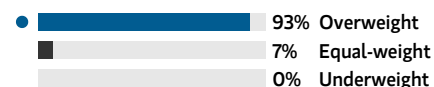
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We expect GigaDevice to continue to gain share in China's MCU market. Auto MCUs offer bigger opportunities as local self-sufficiency is low.
- We see further NOR Flash pricing upside, and expect GigaDevice NOR quality to catch up in the auto segment.
- Specialty DRAM pricing started to rebound in late 1Q25, and there are opportunities such as wafer-on-wafer specialty memory.
- We believe GigaDevice will ramp up 16G LPDDR4 and 32G DDR4 at CXMT's Gen 4 platform in 2027.
- Our price target implies 2026e P/E of 46x, slightly above the historical average of 40x since 2016.

Consensus Rating Distribution



● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Positive*
Secular Growth: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

Rmb1,016.00

79x 2026e EPS

NOR flash prices rise more sharply into 2H26, and the company increases production of mid-/high-density NOR; MCU growth accelerates; DRAM business shows significant contribution: 1) NOR flash sales grow 120%+ Y/Y in 2026 via price hikes and meaningful growth in demand; 2) MCU sales growth significantly exceeds 40% in 2026; 3) contribution from WoW tops expectations in 2026-27e; 4) DDR4 price hike stronger than expected; 5) gross margin above 60% in 2026.

BASE CASE

Rmb585.00

46x 2026e EPS

NOR and DRAM pricing upside in 2026; moderate WoW contribution: 1) NOR flash sales rise 91%+ Y/Y in 2026; 2) MCU grows 28% in 2026; 3) moderate contribution from WoW in 2026-27e; 4) significant DDR4 price hike in 2026; 5) gross margin at ~57% in 2026.

BEAR CASE

Rmb427.00

33x 2026e EPS

NOR Flash price falls sharply into 2H26; MCU faces stronger competition from foreign and local vendors; pause in DRAM business development: 1) NOR flash sales decline in 2026, as ASP starts to drop; 2) MCU sales decrease >10% Y/Y in 2026; 3) contribution from WoW falls below expectations in 2026-27e; 4) DDR4 price declines; 5) gross margin below 35% in 2026.

Risk Reward – GigaDevice Semiconductor Beijing Inc (603986.SS)

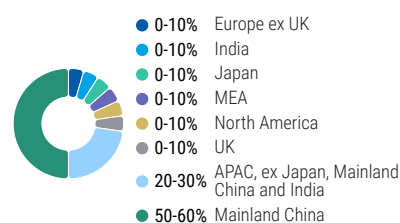
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
NOR Flash ASP growth (%)	4	52	8	0
MCU ASP growth (%)	(8)	0	0	0
Gross margin (%)	40	56	57	54

INVESTMENT DRIVERS

- Beneficiary of MCU localization
- Decreasing opex burden
- Development of AI glasses market

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

4/5
MOST

3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- NOR up-cycle driven by stronger demand
- Superior chip design, leading to increasing exposure to low-density NOR
- Faster-than-expected DRAM development

RISKS TO DOWNSIDE

- NOR down-cycle driven by weaker demand
- Inferior chip design, leading to increasing exposure to mid-/high-density NOR
- Slower-than-expected DRAM development

OWNERSHIP POSITIONING

Inst. Owners, % Active 61.1%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (Rmb, mn) 13,902.0 17,687.8 22,017.7

EBITDA (Rmb, mn) 3,698.0 6,930.3 9,882

Net income (Rmb, mn) 3,396.0 5,919.4 8,589

EPS (Rmb) 5.1 8.5 12.8

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

GigaDevice: Financial Summary

Exhibit 20: Financial Summary

Income Statement						Cash Flow Statement					
Rmb mn (Years End Dec)	2024	2025	2026E	2027E	2028E	Rmb mn (Years End Dec)	2024	2025	2026E	2027E	2028E
Net sales	7,356	9,203	22,018	32,133	37,996	Cashflow from Operations	2,032	2,032	7,544	12,422	14,264
COGS	(4,561)	(5,502)	(9,437)	(13,736)	(17,583)	Net profits	1,103	1,648	8,589	13,881	15,465
Gross profit	2,795	3,701	12,581	18,397	20,413	Depreciation	269	250	250	250	250
Operating expenses	(1,984)	(2,176)	(2,949)	(3,083)	(3,354)	Working Capital Change	(397)	(440)	(1,295)	(1,709)	(1,451)
Operating income	811	1,526	9,632	15,314	17,059	Other adjustments	1,058	574	0	0	0
Non-operating income	313	181	(125)	20	20	Cashflow from investing	(669)	(1,391)	(250)	(250)	(250)
Pre-tax income	1,124	1,706	9,507	15,334	17,079	Capex	(499)	(1,049)	(250)	(250)	(250)
Income tax	23	29	873	1,408	1,568	Change of LT Investment	(1,975)	(1,886)	0	0	0
Minority interests	2	(29)	(45)	(45)	(45)	Change of ST Investment	0	0	0	0	0
Reported net income	1,103	1,648	8,589	13,881	15,465	Other adjustments	1,805	1,544	0	0	0
Adj. wtd. avg. shrs(m)	666	666	670	670	670	Cashflow from financing	480	(517)	(494)	(2,577)	(4,164)
MW EPS (Rmb)	1.66	2.48	12.81	20.71	23.07	Increase in L/T debt	0	0	0	0	0
Reported EPS (Rmb)	1.66	2.48	12.81	20.71	23.07	Increase in S/T debt	898	(698)	0	0	0
						Cash Dividend Paid	(15)	(243)	(494)	(2,577)	(4,164)
						Dir & Emp Bonus Paid	0	0	0	0	0
						Issuance of stock	5	516	0	0	0
						Other adjustments	(407)	(92)	0	0	0
						Exchange rate adjustment	107	21	0	0	0
						Net change in cash	1,950	146	6,800	9,596	9,850
Balance Sheet						Financial Ratios					
Rmb mn (Years End Dec)	2024	2025	2026E	2027E	2028E		2024	2025	2026E	2027E	2028E
Cash	9,128	9,186	15,986	25,582	35,432	Growth(%)					
Mkt Securities	120	102	102	102	102	Turnover	27.7	25.1	139.2	45.9	18.2
AR/NR	232	199	537	784	927	Operating profits	129.9	88.1	531.4	59.0	11.4
Inventory	2,346	3,066	4,487	6,532	8,361	EPS	585.3	49.6	417.5	61.6	11.4
Other	609	873	873	873	873	Margins (%)					
Current Assets	12,435	13,425	21,985	33,872	45,694	Gross Margin	38.0	40.2	57.1	57.3	53.7
Long-term investments	3,503	4,085	4,085	4,085	4,085	Operating Margin	11.0	16.6	43.7	47.7	44.9
Fixed assets	1,062	1,325	1,325	1,325	1,325	Pretax Margin	15.3	18.5	43.2	47.7	44.9
Deferred assets	426	432	432	432	432	Net Margin	15.0	17.9	39.0	43.2	40.7
Other assets	1,803	2,129	2,129	2,129	2,129	Return (%)					
Total Assets	19,229	21,397	29,956	41,843	53,665	ROAE	6.9	9.2	36.9	42.1	34.9
S/T borrowings	898	200	200	200	200	ROAA	6.2	8.1	33.5	38.7	32.4
AP/INP	734	813	1,278	1,860	2,382	Gearing (%)					
Other ST liabilities	699	934	934	934	934	Net Debt/Equity	(49.3)	(46.7)	(57.8)	(65.7)	(70.6)
LT debt	0	0	0	0	0	Liabilities/Equity	15.3	11.3	9.7	8.3	7.5
Other LT liabilities	220	226	226	226	226	Ratios (X)					
Total Liabilities	2,550	2,174	2,639	3,221	3,742	Current ratio	5.3	6.9	9.1	11.3	13.0
Common shares	664	668	668	668	668	Quick ratio	4.0	4.8	6.8	8.8	10.3
Additional capital	8,656	9,201	9,201	9,201	9,201	Others					
Retained earning	6,796	8,407	16,502	27,806	39,107	AR/NR Turnover (days)	9	9	9	9	9
Other shareholders' equity	563	946	946	946	946	Inventory Turnover (days)	174	174	174	174	174
Total Equity	16,679	19,223	27,317	38,622	49,923	AP Turnover (days)	49	49	49	49	49
Total Liab. & Shrhldr's Equity	19,229	21,397	29,956	41,843	53,665	Cash Conversion (days)	133	133	133	133	133

E = Morgan Stanley Research Estimates
Source: Morgan Stanley Research, Company Data

Nanya Tech: Estimate Revisions and Quarterly Financials

We raise our EPS estimates by 15% for 2026, 29% for 2027, and 33% for 2028, reflecting DDR4 pricing strength. We also turn positive on more LTAs with server and enterprise SSD customers, along with their additional 5kwp capacity to DDR5 starting 3Q26..

Exhibit 21: Nanya Tech: Estimate revisions

NT\$ mn	New '26E	Old '26E	Diff.	New '27E	Old '27E	Diff.	New '28E	Old '28E	Diff.
Net sales	332,719	301,785	10%	471,750	377,418	25%	341,113	275,445	24%
Gross profit	258,553	230,616	12%	359,775	276,873	30%	205,545	154,803	33%
Operating profit	241,654	205,800	17%	319,538	242,266	32%	174,123	127,189	37%
Pretax Income	246,334	209,831	17%	323,815	245,628	32%	178,412	130,563	37%
Net income	198,340	169,050	17%	260,940	197,849	32%	144,097	105,458	37%
EPS (NT\$)	61.50	53.71	15%	80.92	62.86	29%	44.68	33.50	33%
Margins									
Gross margin	77.7%	76.4%		76.3%	73.4%		60.3%	56.2%	
Operating margin	72.6%	68.2%		67.7%	64.2%		51.0%	46.2%	
Pretax margin	74.0%	69.5%		68.6%	65.1%		52.3%	47.4%	
Net margin	59.6%	56.0%		55.3%	52.4%		42.2%	38.3%	
BVPS (NT\$)									
	New '26E	Old '26E	Diff.	New '27E	Old '27E	Diff.	New '27E	Old '27E	Diff.
BVPS (NT\$)	117.90	109.59	8%	202.11	173.44	17%	248.61	207.48	20%

Source: Morgan Stanley Research (E) estimates

Exhibit 22: Nanya Tech: Quarterly financials

NT\$ in million	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026e	2027e	2028e
Total Revenues	49,087	79,416	97,208	107,008	123,151	123,242	115,934	109,423	34,132	66,587	332,719	471,750	341,113
Sequential Change	63.1%	61.8%	22.4%	10.1%	15.1%	0.1%	-5.9%	-5.6%					
Change vs Year Ago	582.9%	654.5%	417.6%	255.6%	150.9%	55.2%	19.3%	2.3%	14.2%	95.1%	399.7%	41.8%	-27.7%
Cost of Sales	(15,771)	(17,635)	(19,500)	(21,260)	(22,522)	(23,977)	(25,432)	(40,044)	(32,292)	(50,841)	(74,166)	(111,975)	(135,568)
Percent of Revenues	32.1%	22.2%	20.1%	19.9%	18.3%	19.5%	21.9%	36.6%	94.6%	76.4%	22.3%	23.7%	39.7%
Variable costs	(12,874)	(14,650)	(16,453)	(18,003)	(19,658)	(21,113)	(22,568)	(25,451)	(18,408)	(37,321)	(61,981)	(88,791)	(111,163)
as a % of revenue	26.2%	18.4%	16.9%	16.8%	16.0%	17.1%	19.5%	23.3%	53.9%	56.0%	18.6%	18.8%	32.6%
Dep. & Amort. expense	(2,897)	(2,966)	(3,046)	(3,257)	(2,864)	(2,864)	(2,864)	(14,593)	(13,883)	(13,520)	(12,186)	(23,184)	(24,404)
as a % of COGS	18.4%	16.9%	15.6%	15.3%	12.7%	11.9%	11.3%	36.4%	43.0%	26.6%	16.4%	20.7%	18.0%
as a % of revenue	5.9%	3.8%	3.1%	3.0%	2.3%	2.3%	2.5%	13.3%	40.7%	20.3%	3.7%	4.9%	7.2%
Gross Margin	33,316	61,781	77,708	85,748	100,629	99,265	90,502	69,379	1,840	15,745	258,553	359,775	205,545
Percent of Revenues	67.9%	77.8%	79.9%	80.1%	81.7%	80.5%	78.1%	63.4%	5.4%	23.6%	77.7%	76.3%	60.3%
Total Opex	(3,205)	(3,987)	(4,592)	(5,114)	(10,066)	(10,211)	(10,043)	(9,916)	(10,134)	(9,741)	(16,899)	(40,236)	(31,422)
Percent of Revenues	6.5%	5.0%	4.7%	4.8%	8.2%	8.3%	8.7%	9.1%	29.7%	14.6%	5.1%	8.5%	9.2%
R&D	(2,109)	(2,214)	(2,325)	(2,511)	(2,800)	(2,940)	(3,087)	(3,241)	(7,685)	(7,041)	(9,160)	(12,068)	(11,637)
Percent of Revenues	4.3%	2.8%	2.4%	2.3%	2.3%	2.4%	2.7%	3.0%	22.5%	10.6%	2.8%	2.6%	3.4%
Sales and Marketing	(347)	(562)	(688)	(757)	(1,970)	(1,972)	(1,855)	(1,751)	(665)	(832)	(2,354)	(7,548)	(5,458)
Percent of Revenues	0.7%	0.7%	0.7%	0.7%	1.6%	1.6%	1.6%	1.6%	1.9%	1.2%	0.7%	1.6%	1.6%
General and Admin	(748)	(1,211)	(1,579)	(1,846)	(5,295)	(5,299)	(5,101)	(4,924)	(1,784)	(1,868)	(5,385)	(20,620)	(14,327)
Percent of Revenues	1.5%	1.5%	1.6%	1.7%	4.3%	4.3%	4.4%	4.5%	5.2%	2.8%	1.6%	4.4%	4.2%
Operating Income	30,111	57,794	73,115	80,634	90,563	89,054	80,459	59,463	(8,294)	6,004	241,654	319,538	174,123
Percent of Revenues	61.3%	72.8%	75.2%	75.4%	73.5%	72.3%	69.4%	54.3%	-24.3%	9.0%	72.6%	67.7%	51.0%
EBITDA	33,008	60,779	76,162	83,891	93,426	91,918	83,322	74,056	5,589	19,524	253,840	342,723	198,528
EBITDA margin	67%	77%	78%	78%	76%	75%	72%	68%	16%	29%	76%	73%	58%
Non-operating Income(Loss)	1,607	1,020	1,020	1,034	1,070	1,070	1,070	1,067	3,998	2,639	4,680	4,277	4,288
Profit Before Taxes	31,718	58,813	74,135	81,667	91,633	90,124	81,529	60,530	(4,297)	8,643	246,334	323,815	178,412
Percent of Revenues	64.6%	74.1%	76.3%	76.3%	74.4%	73.1%	70.3%	55.3%	-12.6%	13.0%	74.0%	68.6%	52.3%
Taxes	(5,060)	(11,512)	(16,843)	(13,979)	(16,351)	(17,640)	(18,523)	(10,361)	1,474	(1,269)	(47,994)	(62,875)	(34,314)
Tax Rate	17.8%	19.6%	22.7%	17.1%	17.8%	19.6%	22.7%	17.1%	34.3%	14.7%	19.5%	19.4%	19.2%
Reported Income (TW GAAP)	26,058	47,302	57,292	67,688	75,281	72,484	63,006	50,169	(2,823)	7,373	198,340	260,940	144,097
Percent of Revenues	53.1%	59.6%	58.9%	63.3%	61.1%	58.8%	54.3%	45.8%	-8.3%	11.1%	59.6%	55.3%	42.2%
Change vs Year Ago	-144.3%	-125.3%	356.5%	510%	189%	53%	10%	-26%	-59%	-361%	2590%	32%	-45%
Reported EPS (NT\$, TW GAAP)	8.08	14.67	17.77	20.99	23.34	22.48	19.54	15.56	(0.91)	2.36	61.50	80.92	44.68
Sequential Change	129%	82%	21%	18%	11%	-4%	-13%	-20%	-59%	-359%	2505%	32%	-45%
Book value per share									53.3	54.6	113.3	194.2	238.9

Source: Company data, Morgan Stanley Research (E) estimates

Nanya Tech: Valuation Methodology

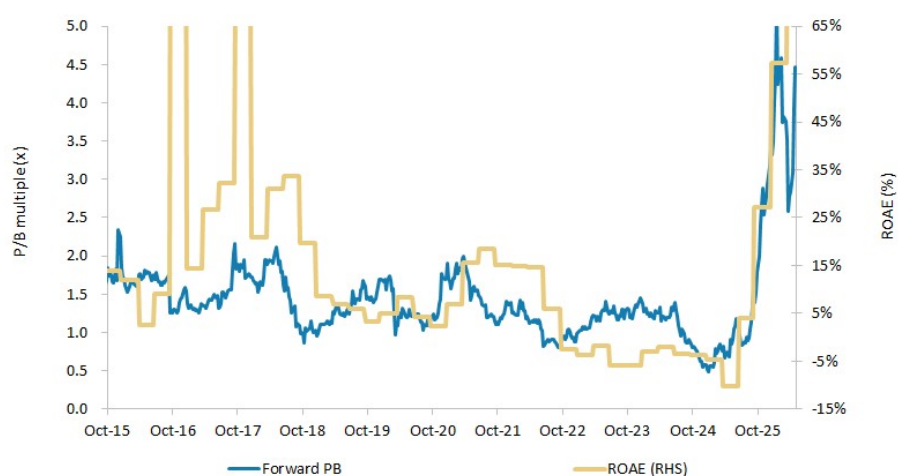
We raise our price target (base case scenario value) from NT\$278 to NT\$380. Our price target reflects BVPS estimates of 3.22x 2026 and 1.88x 2027, vs. ROE seen at 74% for 2026 and 53% for 2027. We expect DDR4 pricing and strength plus LTAs to drive re-rating. We raise our bull and bear case values from NT\$588 and NT\$155 to NT\$805 and NT\$210, respectively, implying 6.83x and 1.78x 2026e BVPS.

Exhibit 23: Nanya Tech – Historical P/B band

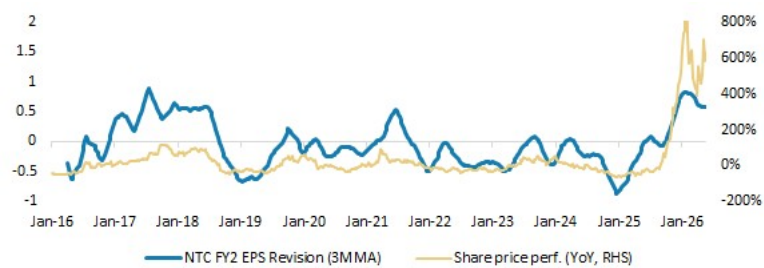


Source: Company data, Morgan Stanley Research estimates

Exhibit 24: Nanya Tech: Historical forward P/B vs ROE



Source: Company data, Morgan Stanley Research estimates

Exhibit 25: Nanya Tech: Earnings estimate revision breadth

Source: Company data, Morgan Stanley Research estimates

Risk Reward – Nanya Technology Corp. (2408.TW)

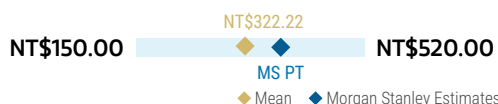
DDR4 S/D becoming more favorable; OW

PRICE TARGET NT\$380.00

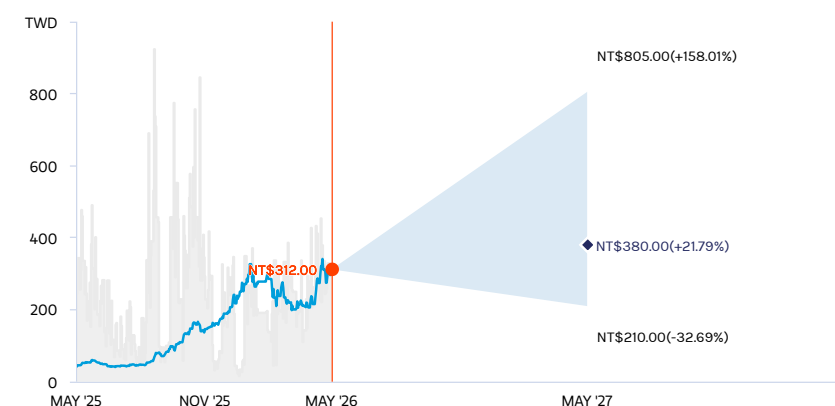
Base case, P/B. We expect the stock to trade at 3.22x 2026, 1.88x 2027, and 1.53x 2028 BVPS, above its average since 2015. We think NTC will benefit from a supply-driven up-cycle as major memory players are exiting the DDR4 market. That should help NTC more than offset the competitive impact from CXMT in the near term.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



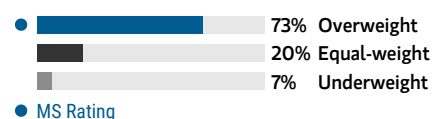
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- Major memory suppliers are gradually exiting the DDR4 market, benefiting existing Taiwanese players, including Nanya Tech.
- We expect DDR4 shortage into 2H26 to help NTC more than offset the competitive impact from CXMT in 2026. The supply shortage appears set to widen into 2H26.
- Our price target reflects BVPS of 3.22x for 2026, 1.88x for 2027, and 1.53x for 2028.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

NT\$805.00

6.83x 2026e BVPS

DRAM prices strong; faster-than-expected 1a/1b nm ramp; competition from Korea and China is a non-factor: DRAM pricing becomes very strong in 2026 thanks to production cuts and relief of inventory pressure. NTC's utilization rate returns to the 100% level. Korean and Chinese competitors all exit the consumer DRAM market.

BASE CASE

NT\$380.00

3.22x 2026e BVPS

Strong DRAM cycle: Major DRAM players now plan to exit the DDR4 market, which positions Taiwanese players to benefit. We expect price-hike momentum to continue through 2026, with widened supply shortage into 2H26.

BEAR CASE

NT\$210.00

1.78x 2026e BVPS

Specialty DRAM demand becomes weak in 2026 with intensifying Chinese competition: Nanya Tech could face more pressure from pricing and technological competition amid a weak cycle. Inventory remains at historical peak levels. SK hynix and Chinese DRAM competitors take all the market share in China. Nanya Tech's utilization rate falls to the 40% level.

Risk Reward – Nanya Technology Corp. (2408.TW)

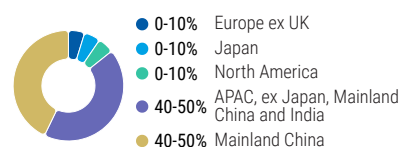
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
12" Wafers Out per quarter (000s)	825.0	830.0	855.0	900.0
Average Wafer Yield (%)	81.8	108.1	106.5	95.0
Total Die shipments - 2 gb equivalent (mn)	2,368.0	3,178.8	3,244.9	3,090.4

INVESTMENT DRIVERS

- Specialty DRAM pricing
- Monthly sales momentum
- Industry outlook reported by industry peers

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

2/5
MOST

3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Sustained DRAM prices from more disciplined supply and stronger demand
- Stronger pricing environment, enabling revenue guidance to be beaten
- Faster-than-expected 1a/1b nm ramp-up progress

RISKS TO DOWNSIDE

- Slower-than-expected 1a/1b nm ramp-up
- Less demand for specialty DRAM from 4K2K TVs and smart set-top boxes

OWNERSHIP POSITIONING

Inst. Owners, % Active 54.7%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (NT\$, mn) 259,718 294,571 332,719

EBITDA (NT\$, mn) 191,966 223,622 253,840 254,943

Net income (NT\$, mn) 154,740 173,479 198,340 198,615

EPS (NT\$) 40.62 49.69 61.50

◆ Mean ◆ Morgan Stanley Estimates
Source: Refinitiv, Morgan Stanley Research

Nanya Tech: Financial Summary

Income Statement

NT\$m (Years End Dec)	2024	2025	2026E	2027E	2028E
Net sales	34,132	66,587	332,719	471,750	341,113
COGS	(32,292)	(50,841)	(74,166)	(111,975)	(135,568)
Depreciation Expense	(13,883)	(13,520)	(12,186)	(23,184)	(24,404)
Variable Cost	(18,408)	(37,321)	(61,981)	(88,791)	(111,163)
Gross profit	1,840	15,745	258,553	359,775	205,545
Operating expenses	(10,134)	(9,741)	(16,899)	(40,236)	(31,422)
Operating income	(8,294)	6,004	241,654	319,538	174,123
Non-operating income	3,998	2,639	4,680	4,277	4,288
Pre-tax income	(4,297)	8,643	246,334	323,815	178,412
Income tax	1,474	(1,269)	(47,994)	(62,875)	(34,314)
Reported net income	(2,823)	7,373	198,340	260,940	144,097
Adj. wtd. avg. shrs (m)	3,099	3,123	3,225	3,225	3,225
Reported EPS (NT\$)	(0.91)	2.36	61.50	80.92	44.68
EPS for consensus (NT\$)	(0.91)	2.36	61.50	80.92	44.68

Balance Sheet

NT\$m (Years End Dec)	2024	2025	2026E	2027E	2028E
Cash	61,903	58,074	73,074	328,871	603,415
Mkt Securities	0	0	0	0	0
AR/NR	4,132	16,558	54,623	55,855	36,120
Inventory	35,318	27,288	371,653	380,041	245,759
Other	6,593	6,621	16,982	17,264	12,739
Current Assets	107,946	108,541	516,331	782,031	898,033
Long-term investments	4,645	5,904	7,583	8,538	9,492
Fixed assets	84,327	85,031	125,252	132,068	137,663
Other assets	9,789	8,977	5,145	5,145	5,145
Total Assets	206,706	208,453	654,311	927,781	1,050,333
S/T borrowings	14,536	5,060	240,000	245,000	225,000
AP/INP	5,180	6,481	8,496	16,002	14,462
Other ST liabilities	6,330	7,480	17,772	17,796	17,791
LT debt	3,995	14,196	17,796	17,796	17,796
Other LT liabilities	4,878	4,698	4,927	4,927	4,927
Total Liabilities	41,653	37,914	288,991	301,522	279,977
Common stocks	30,986	30,986	30,986	30,986	30,986
Preferred stocks	0	0	0	0	0
Capital Reserve	32,834	33,081	33,168	33,168	33,168
Retained earning	101,233	106,333	301,028	561,968	706,065
Other shareholders' equity	0	0	0	0	0
Total Equity	165,053	170,538	365,319	626,259	770,357
Total Liab. & Shrhldr's Equity	206,706	208,453	654,311	927,781	1,050,333
BVPS	53.27	54.61	113.28	194.20	238.89

Cash Flow Statement

NT\$m (Years End Dec)	2024	2025	2026E	2027E	2028E
Cashflow from Operations	1,952	18,568	(174,468)	280,797	324,544
Net profits	(2,823)	7,373	198,340	260,940	144,097
Depreciation	15,892	14,034	12,186	23,184	24,404
Working Capital Change	8,316	5,697	(380,415)	(2,114)	152,477
Other adjustments	(19,433)	(8,536)	(4,579)	(1,213)	3,566
Cashflow from Investing	(16,188)	(14,110)	(50,005)	(30,000)	(30,000)
Capex	(16,141)	(13,441)	(50,000)	(30,000)	(30,000)
Change of LT Investment	0	(612)	0	0	0
Change of ST Investment	0	0	0	0	0
Other adjustments	(46)	(57)	(5)	0	0
Cashflow from financing	13,707	(6,317)	238,453	5,000	(20,000)
Increase in L/T debt	6,750	(6,750)	3,600	0	0
Increase in S/T debt	3,355	(9,477)	234,941	5,000	(20,000)
Cash Dividend Paid	0	0	0	0	0
Dir & Emp Bonus Paid	0	(7,000)	0	0	0
Proceed fm New Issue	0	0	0	0	0
Other adjustments	3,602	16,909	(87)	0	0
Exchange rate adjustment	3,619	(1,970)	1,019	0	0
Net change in cash	3,091	-3,829	15,000	255,797	274,544

Financial Ratios

	2024	2025	2026E	2027E	2028E
Growth (%)					
Turnover	14.2	95.1	399.7	41.8	-27.7
Margins (%)					
Gross Margin	5.4	23.6	77.7	76.3	60.3
Operating Margin	-24.3	9.0	72.6	67.7	51.0
EBITA Margin	-65.0	-11.3	69.0	62.8	43.9
Pretax Margin	-12.6	13.0	74.0	68.6	52.3
Net Profit	-8.3	11.1	59.6	55.3	42.2
Return (%)					
ROAE	(1.7)	4.4	74.0	52.6	20.6
ROAA	(1.4)	3.6	46.0	33.0	14.6
Gearing (%)					
Net Debt/Equity	(26.3)	(22.7)	50.6	(10.6)	(46.8)
Liabilities/Equity	25.2	22.2	79.1	48.1	36.3
Ratios (X)					
Current ratio	4.1	5.7	1.9	2.8	3.5
Quick ratio	2.5	3.9	0.5	1.4	2.5
Others					
AR/NR Turnover (days)	59	50	57	39	43
Inventory Turnover (days)	311	337	172	219	291
AP Turnover (days)	37	59	47	42	52
Cash Conversion (days)	333	328	182	216	281

Source: Company data, TEJ, Morgan Stanley ResearchE= Morgan Stanley Research Estimates

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Risk Reward Reference links

1. View explanation of Options Probabilities methodology - [Options_Probabilities_Exhibit_Link.pdf](#)
2. View descriptions of Risk Rewards Themes - [RR_Themes_Exhibit_Link.pdf](#)
3. View explanation of regional hierarchies - [GEG_Exhibit_Link.pdf](#)
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Equal-weight/Hold	1568	43%	358	39%	23%	715	44%
Not-Rated/Hold	4	0%	0	0%	0%	1	0%
Underweight/Sell	555	15%	84	9%	15%	202	12%
Total	3,673		909			1627	

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INDUSTRY COVERAGE: Greater China Technology Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (05/27/2026)
Charlie Chan		
ACM Research Inc (ACMR.O)	O (03/07/2023)	US\$88.64
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb455.05
Advanced Wireless Semiconductor Co (8086.TWO)	U (07/14/2025)	NT\$173.50
Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT\$4,500.00
ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT\$642.00
Cambricon Technology Corporation (688256.SS)	O (04/27/2026)	Rmb1,340.00
Global Unichip Corp (3443.TW)	O (07/27/2024)	NT\$5,100.00
GlobalWafers Co Ltd (6488.TWO)	E (05/19/2026)	NT\$930.00
Gudeng Precision (3680.TWO)	O (11/25/2025)	NT\$557.00

Hua Hong Semiconductor Ltd (1347.HK)	E (03/12/2026)	HK\$152.40
Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)	O (04/27/2026)	HK\$536.50
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT\$315.50
Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb116.97
MediaTek (2454.TW)	O (11/28/2025)	NT\$4,640.00
MetaX Integrated Circuits (688802.SS)	E (04/27/2026)	Rmb695.00
Nanya Technology Corp. (2408.TW)	O (05/28/2026)	NT\$312.00
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb654.77
OmniVision Integrated Circuits Group Inc (603501.SS)	E (11/17/2025)	Rmb104.96
Phison Electronics Corp (8299.TWO)	E (02/25/2026)	NT\$2,480.00
SG Micro Corp. (300661.SZ)	E (11/03/2025)	Rmb119.39
Silergy Corp. (6415.TW)	U (05/19/2026)	NT\$665.00
SMIC (0981.HK)	O (10/21/2025)	HK\$85.20
TSMC (2330.TW)	O (02/07/2022)	NT\$2,300.00
UMC (2303.TW)	O (05/19/2026)	NT\$143.50
Vanguard International Semiconductor (5347.TWO)	E (01/14/2026)	NT\$166.00
WIN Semiconductors Corp (3105.TWO)	U (07/14/2025)	NT\$560.00
Daisy Dai, CFA		
ASMP Ltd (0522.HK)	O (07/24/2025)	HK\$203.80
China Resources Microelectronics Limited (688396.SS)	U (03/02/2026)	Rmb65.98
Elan Microelectronics Corp (2458.TW)	O (10/03/2025)	NT\$164.50
Empyrean Technology Co Ltd (301269.SZ)	E (01/17/2025)	Rmb127.42
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	U (08/25/2025)	Rmb35.58
Innoscence (2577.HK)	E (10/13/2025)	HK\$80.30
JCET Group Co Ltd (600584.SS)	E (01/16/2026)	Rmb86.16
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK\$36.94
SICC Co Ltd (688234.SS)	O (03/20/2026)	Rmb166.99
StarPower Semiconductor Ltd (603290.SS)	E (05/14/2026)	Rmb135.00
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb85.06
Universal Scientific Ind. (Shanghai) (601231.SS)	O (11/05/2025)	Rmb43.43
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb110.02
Daniel Yen, CFA		
AP Memory Technology Corp (6531.TW)	O (07/11/2025)	NT\$1,080.00
ASMedia Technology Inc (5269.TW)	U (10/03/2025)	NT\$1,495.00
Aspeed Technology (5274.TWO)	O (06/09/2025)	NT\$18,230.00
Egis Technology Inc (6462.TWO)	E (01/28/2026)	NT\$133.00
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb181.08
GigaDevice Semiconductor Beijing Inc (603986.SS)	O (05/15/2025)	Rmb514.18
Macronix International Co Ltd (2337.TW)	O (09/18/2025)	NT\$155.50
Montage Technology Co Ltd (6809.HK)	O (03/18/2026)	HK\$469.00
Montage Technology Co Ltd (688008.SS)	O (03/18/2026)	Rmb265.50
Novatek (3034.TW)	U (02/04/2026)	NT\$485.00
Nuvoton Technology Corporation (4919.TW)	U (11/10/2025)	NT\$206.50
Parade Technologies Ltd (4966.TWO)	O (05/27/2026)	NT\$870.00
Powerchip Semiconductor Manufacturing Co (6770.TW)	O (10/27/2025)	NT\$74.80
Realtek Semiconductor (2379.TW)	E (01/30/2026)	NT\$605.00
Shenzhen Goodix Technology Co Ltd (603160.SS)	U (07/14/2025)	Rmb66.50
Winbond Electronics Corp (2344.TW)	O (05/28/2026)	NT\$155.00
WPG Holdings (3702.TW)	O (03/16/2026)	NT\$120.50
WT Microelectronics Co. Ltd. (3036.TW)	O (01/27/2026)	NT\$296.00
Duan Liu		
Dosilicon Co Ltd (688110.SS)	U (09/06/2024)	Rmb148.00
Shenzhen Longsys Electronics Co Ltd (301308.SZ)	E (02/25/2026)	Rmb551.59
Tiffany Yeh		

AllRing Tech Co. (6187.TWO)	O (09/23/2025)	NT\$1,130.00
FOCI Fiber Optic Communications Inc (3363.TWO)	O (01/15/2025)	NT\$847.00
Himax Technologies Inc (HIMX.O)	E (02/04/2026)	US\$20.65
Hon Precision (7769.TW)	O (04/17/2026)	NT\$7,800.00
MPI Corporation (6223.TWO)	O (04/17/2026)	NT\$6,390.00
Silicon Motion (SIMO.O)	O (05/06/2024)	US\$284.98
WinWay Technology Co Ltd (6515.TW)	O (04/17/2026)	NT\$9,795.00

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* Historical prices are not split adjusted.

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